

# pdf-studio

A thin, type-safe wrapper around ReportLab for developers who need real PDFs — tables, charts, headers, KPIs — without 300 lines of boilerplate. This document was generated by pdf-studio itself.

## What's Inside

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- Three research-backed themes (cypher / ledger / slate) — WCAG-AA verified
- Declarative templates: data → styled PDF in one call
- Real bold/italic via bundled Inter, Lora, JetBrains Mono TTFs
- Vector charts (matplotlib → SVG → PDF) — crisp at any zoom
- Styled tables with zebra striping, right-aligned numerics, captions
- KPI summary cards with green/red deltas
- Running headers with {page} / {total} placeholders
- Zero required dependencies beyond reportlab

# Theme Palettes

Each theme is a complete visual language. Switching themes recolors headings, tables, charts, KPIs, bullets — no code changes.

## Cypher Theme

Navy foundation, teal accent. The brand default.

cypher colour roles

| Role            | Hex     | Preview |
|-----------------|---------|---------|
| foundation      | #0B1121 | ■       |
| accent (fills)  | #2DD4BF | ■       |
| good (up)       | #047857 | ■       |
| bad (down)      | #B91C1C | ■       |
| surface (cards) | #F8FAFC | ■       |
| body text       | #1F2937 | ■       |
| muted text      | #64748B | ■       |
| grid lines      | #E2E8F0 | ■       |

## Ledger Theme

Deep-green foundation, gold accent. Finance-optimised.

ledger colour roles

| Role            | Hex     | Preview |
|-----------------|---------|---------|
| foundation      | #064E3B | ■       |
| accent (fills)  | #B45309 | ■       |
| good (up)       | #047857 | ■       |
| bad (down)      | #B91C1C | ■       |
| surface (cards) | #F0FDF4 | ■       |
| body text       | #1F2937 | ■       |
| muted text      | #374151 | ■       |
| grid lines      | #D1FAE5 | ■       |

# Slate Theme

Indigo foundation, amber accent. Contemporary, warm.

slate colour roles

| Role            | Hex     | Preview |
|-----------------|---------|---------|
| foundation      | #312E81 | ■       |
| accent (fills)  | #D97706 | ■       |
| good (up)       | #047857 | ■       |
| bad (down)      | #B91C1C | ■       |
| surface (cards) | #EEF2FF | ■       |
| body text       | #334155 | ■       |
| muted text      | #475569 | ■       |
| grid lines      | #E0E7FF | ■       |

# Typography & Components

## Heading Hierarchy

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### Level 1 — Section Heading

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### Level 2 — Subsection

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### Level 3 — Detail

Body text uses the theme's `body_text` colour (Inter Regular). Bold and italic resolve to real bundled TTF weights — Inter-Bold, Lora-Italic, JetBrainsMono-Bold. No synthetic rendering.

## KPI Summary Cards

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| AUM     | Realised P&L; | Max Drawdown | Sharpe Ratio |
|---------|---------------|--------------|--------------|
| ₹1.24 L | ₹8.3 K        | -6.2%        | 1.84         |
| +4.1%   | +12.4%        | -0.8%        | +0.12        |

## Feature Bullets

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- Theme dataclass with 3 presets — one object, entire document recolored
- `Document.from_template()` — structured data → styled PDF in one call
- Charts render as vector SVG → PDF (crisp at any zoom level)
- Tables from pandas DataFrame or list-of-lists with right-aligned columns
- Thread-safe font registration with double-checked locking

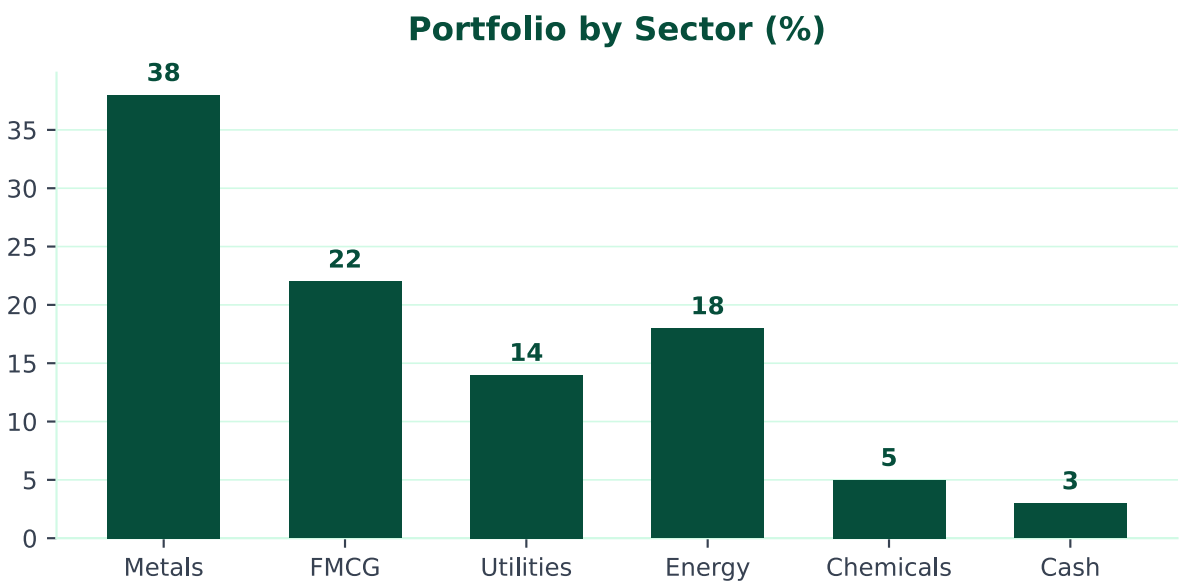
# Tables & Standard Charts

## Holdings Table

Table 1 — Portfolio positions (numeric columns right-aligned)

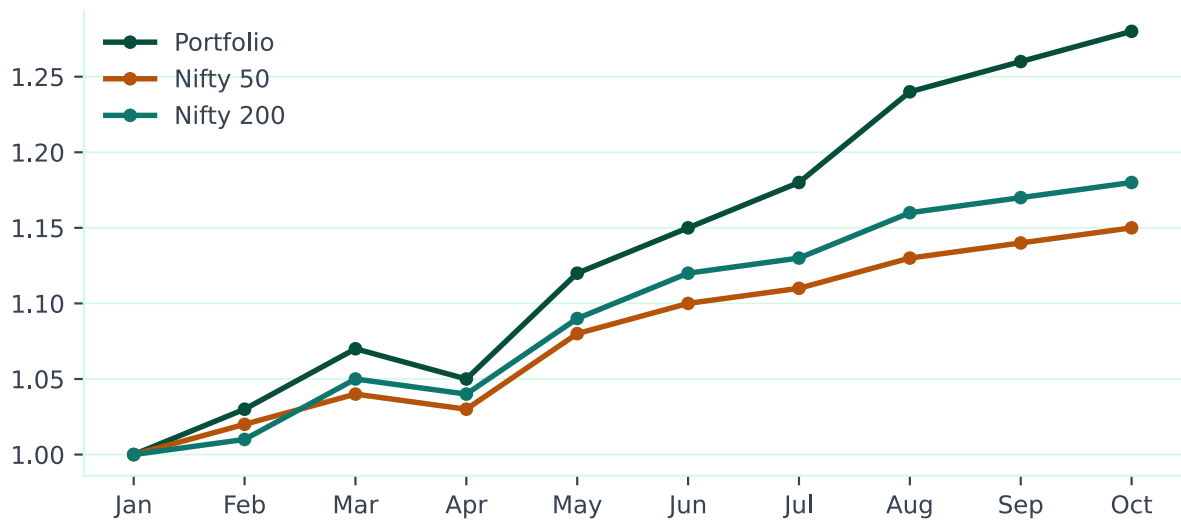
| Holding      | Sector    | Qty | Avg Cost | LTP    | Value   | P&L;   | P&L; % |
|--------------|-----------|-----|----------|--------|---------|--------|--------|
| VEDL         | Metals    | 50  | ₹285     | ₹312   | ₹15,600 | ₹1,350 | +9.5%  |
| ITC          | FMCG      | 20  | ₹412     | ₹445   | ₹8,900  | ₹660   | +8.0%  |
| POWERGRID    | Utilities | 15  | ₹198     | ₹224   | ₹3,360  | ₹390   | +13.1% |
| NMDC         | Metals    | 50  | ₹156     | ₹172   | ₹8,600  | ₹800   | +10.3% |
| CASTROLINDIA | Energy    | 30  | ₹134     | ₹142   | ₹4,260  | ₹240   | +6.0%  |
| SRF          | Chemicals | 4   | ₹2,450   | ₹2,620 | ₹10,480 | ₹680   | +6.9%  |

## Bar Chart — Sector Allocation



## Line Chart — Equity Curve vs Benchmark

### Normalised Returns (Jan = 1.00)

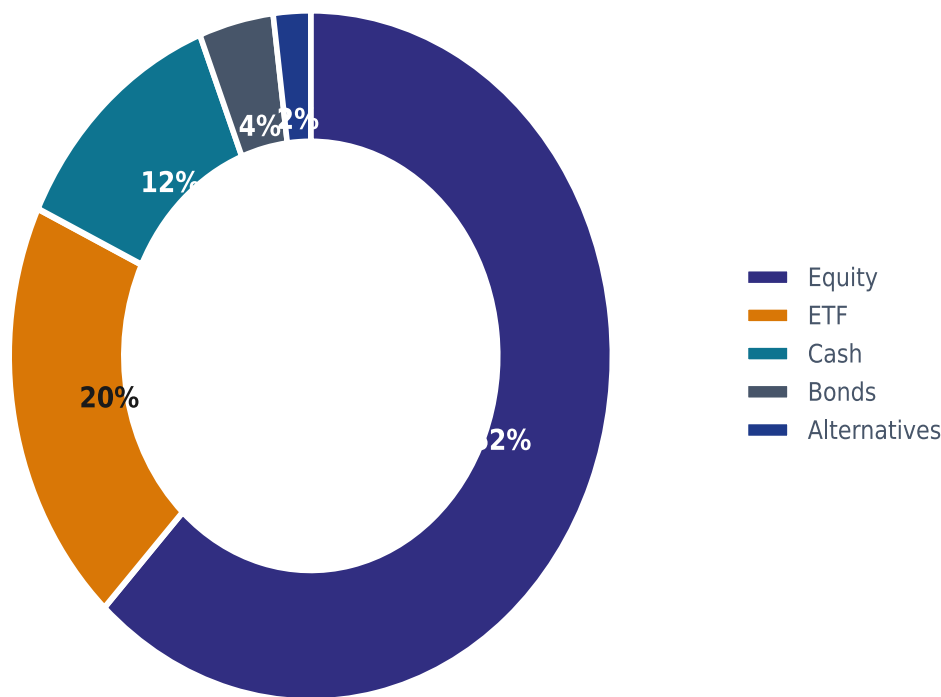


# Composition & Correlation

## Donut Chart — Asset Composition

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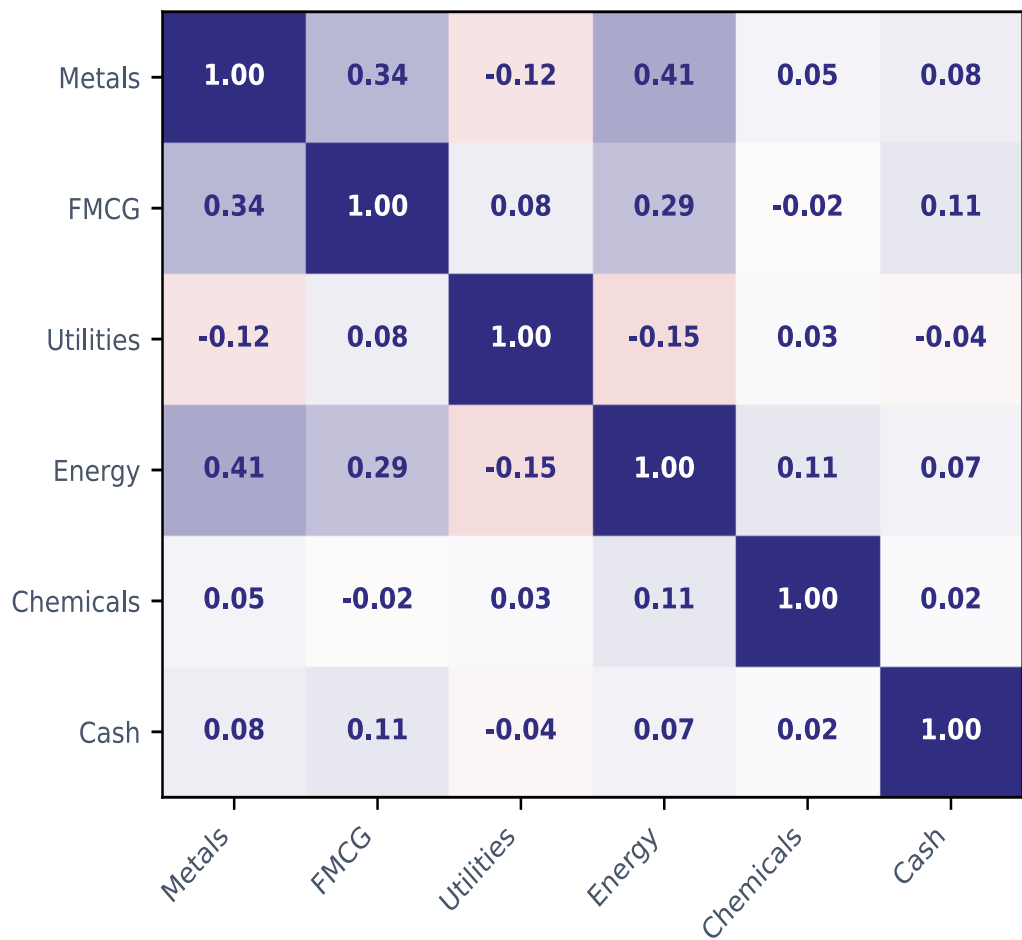
Allocation Breakdown



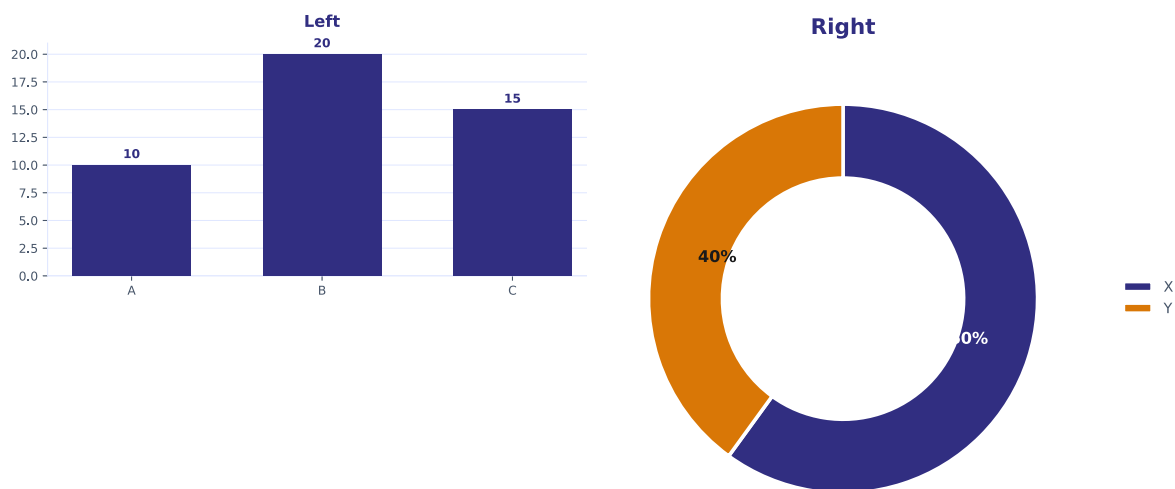
## Heatmap — Sector Correlation Matrix

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## Cross-Sector Correlation



## Chart Row — Two Charts Side-by-Side



# Declarative Template

The `financial_statement` template builds a complete report from structured data — no builder calls needed. One line of code.

## Input Data Structure

Template input schema

| Key              | Type              | Example                                                   |
|------------------|-------------------|-----------------------------------------------------------|
| title            | str               | "Q3 Portfolio Statement"                                  |
| subtitle         | str               | "Generated via<br>from_template()"                        |
| kpis             | list[dict]        | [{"label": "AUM", "value":<br>"₹1.2L", "delta": "+4.1%"}] |
| composition      | tuple[list, list] | (["Equity", "Cash"], [70, 30])                            |
| table            | list[list]        | [["Holding", "Value"], ["VEDL",<br>"₹15,600"]]            |
| right_align_cols | list[int]         | [1]                                                       |

## Usage

```
```python from pdf_studio import Document data = { "title": "Q3 Portfolio", "kpis": [{"label": "AUM",  
"value": "₹1.2L", "delta": "+4.1%"}], "composition": (["Equity", "Cash"], [70, 30]), "table":  
[["Holding", "Value"], ["VEDL", "₹15,600"]], "right_align_cols": [1], } doc =  
Document.from_template("financial_statement", data) doc.render("q3.pdf") ```
```

## That's It

Three themes. One template. Zero boilerplate. `pip install pdf-studio` and start generating.

[github.com/AshayK003/pdf-studio](https://github.com/AshayK003/pdf-studio) | AGPL v3